

A Low-cost Arm Based Motion Restriction Haptics for VR

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ABSTRACT

Virtual reality requires high-level haptics in order to achieve a feeling of realism. In this paper, we present a low-cost device that can provide motion restriction haptics. The device is a wearable, arm based flexible exoskeleton made using cable/zip ties, which are actuated based on VR interaction. It restricts arm motion beyond a point using the cable tie's tensile strength, analogous to how our arm works.

Keywords: Virtual Reality; Haptics; Motion Restriction; Exoskeleton; Cable ties.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction devices—Haptic devices;

1 INTRODUCTION

The sense of proprioception may quickly fall out of sync when interacting with objects in VR. In simple terms. There is no motion restriction to provide the exact sense and presence of virtual objects/obstacles, which is essential since interactions are rich and direct in VR. Hence haptics in VR plays a crucial role in increasing realism [6]. There are also other limitations in the implementation of motion restriction. An equal amount of reaction force is required to stop the arm. To generate such magnitudes of force, heavy motors are to be mounted on the arm, which is not comfortable in VR. This paper describes the low-cost device built using cable/zip tie, servo motor, under 900 INR (12 US Dollar), which provides the essential motion restriction in the VR (see Fig. 1) and then it was tested using interactions designed in Virtual Reality.

2 RELATED WORKS

With an increasing demand for Haptic technology, numerous research has been done to explore haptics. Very close works related to this paper are a new force-feedback arm [4] is a heavy mechanical exoskeleton for haptic interaction in VR. Wireality [3] string-based haptics for tangible geometries in VR. Frozen Suit [2] uses a changeable stiffness suit to provide haptics in-game. An elastic-Arm [1] provides passive haptic feedback for augmenting interaction and perception in VR. Rich and sophisticated haptic devices are primarily associated with labs and are not easily commercialized. This paper presents a low-cost arm-based wearable, flexible exoskeleton, which provides motion restriction haptics in VR.

3 IMPLEMENTATION

An Inverse Kinematics rig was used with the controller as a target to achieve Synchronous Arm motion in VR. After studying the human arm motion and constraints, only elbow and shoulder joints that control the whole limb are considered. The whole arm is connected with several links. Each link is made of cable ties. A link has two ends, One end is fixed on the hinge joint, and another end is attached with rubber bands to provide a smooth retreat. The cable

moves freely through the slot provided in the casing, which has servo motors for actuation.

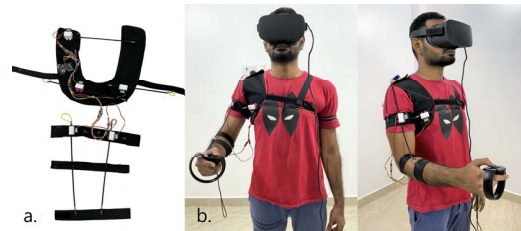


Figure 1: a. Prototype image and b. User wearing the prototype

3.1 Materials

Restraining arm motion in the direction of action after a collision with an obstacle requires a large amount of tensile strength. So Cable ties were preferred, which lock only in one direction. Nylon cable ties are too flexible for operation, so stainless steel cable ties were adhered to it to make it rigid enough.

Electronic components: Arduino Uno WiFi and Micro Servos SG90. **Mechanical components:** cable/zip ties (Steel and nylon), Velcro, rubber bands, 3D-Printing: Solid-works, Interactions built in Unity engine using XR tool kit.

3.2 Weight

One of the main concerns is weight in any worn system [5]. Each module design weighs less than 20g (Weight of motor: 9gm + cable ties: 8g max), so a total of five modules weigh less than 150g. Which is significantly less and so the user does not feel like he is wearing the device.

3.3 Operation Speed And Tracking

Micro servos SG 90 motor has high operation speed in terms of 0.1s/60°, and the designed operating angle is within 15 degrees, so there is very negligible latency. Inside the VR, each module's positions are tagged on the player's armature rig, and relative distance is constantly monitored, which is taken care of by the game engine.

3.4 Core Mechanics

When a cable tie link gets locked, the arm can not be extended beyond that point. Minimal vertical force is enough to lock the cable tie using the cable tie's head. Micro servos SG 90 motor provides 0.16cm/kg force, and it is compact in size (21.5 x 11.8 x 22.7mm). The whole casing was made in 3d print. The casing provides rigid support for the motor, slots, cable ties, and mounted on velcro bands.

Servo motors are Arduino operated, which sends lock, unlock, and intermediate state signals to the motor according to the VR interaction. Lock state: it completely restricts the cable ties movement in only one direction, unlock state: the cable ties can move freely in both the direction (see Fig. 2) Intermediate state: cable locking head is adjusted between the lock and unlock state to provide variable resistance. This provides an opportunity to feel the resistance force when cutting objects in VR.

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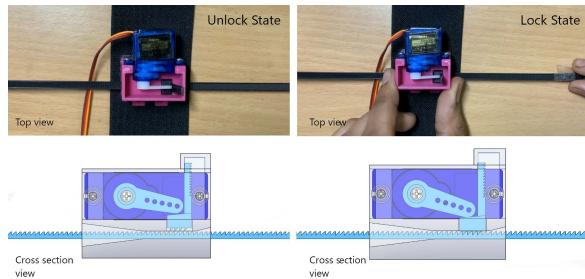


Figure 2: Components of the working module. Unlocked state (left image), locked state (right image)

3.5 Working



Figure 3: The system communicates with the device, the arm is locked, and the motion being arrested when hitting the drums

The player's fist or the tool/weapon triggers the motors to lock the links on any VR collisions. If both the nodes corresponding to the link between the forearm and biceps are moving away and the player touches any obstacles, the game engine sends the signal to a servo motor to lock that respective node. This makes the player unable to move their arm any further, but he can retreat his arm, (see Fig. 3). Similarly, all the links and joints are programmed in the game engine. To demonstrate the prototype's working, following interactions were built-in unity game engines.

4 EVALUATION

Five types of interaction were designed and explored in a VR environment using the prototype, (see Fig. 4). **Drumming action:** demonstrates the collisions with a tool, interaction uses the forearm and elbow movements. **Lifting a sword:** demonstrates the sense of inertia and gravity of the object. **Short duel with the sword:** The player defeats an enemy wielding a sword and shield to demonstrate the interaction of the shoulder movement. **Pressing a button inside the restricted box:** Box has an obstacle on four sides; this demonstrates the touch and motion restriction in the lateral directions. **Punching a bag:** demonstrates sudden recoil haptic feedback.

5 LIMITATIONS AND FUTURE WORK

The prototype does not include any finger-based motion restrictions. It predominantly focuses on forearm and shoulder movement. The concept and prototype were tested on two users so far and planning to conduct more tests on people with different physiques and proportions.

6 CONCLUSION

The prototype's whole build is very cheap, which was made under 900 INR (12 US Dollar), it can be easily made into a commercialized

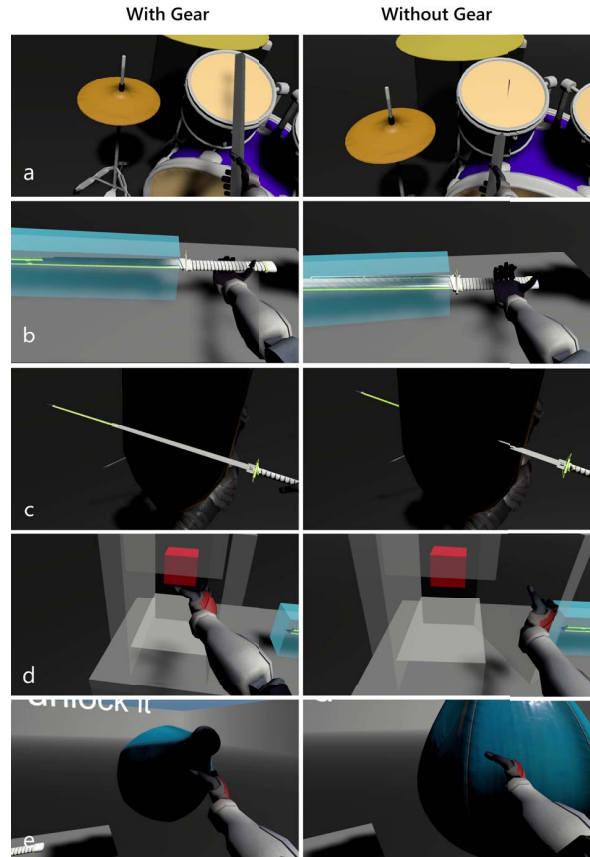


Figure 4: Comparison of interactions with and without prototype. a. Drumming action, b. Lifting a sword, c. Short duel with sword, d. Pressing a button in a restricted box, e. Punching a bag.

product, or an opportunity for DIY model provided the 3d print file, considering the build and cost.

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